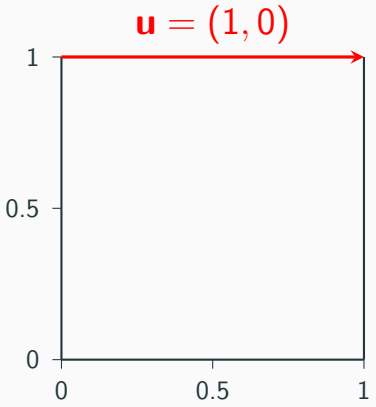
Stokes flow in a lid-driven cavity

Lukas Lanik

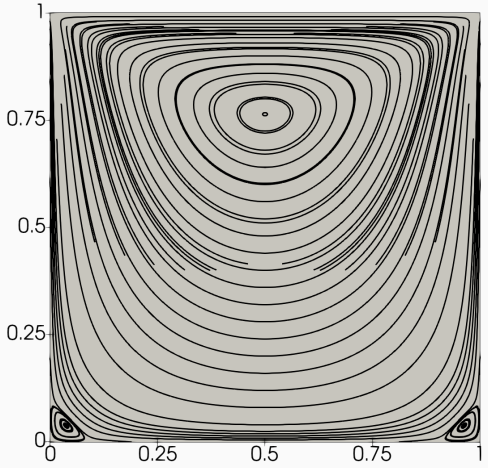
October 13, 2026

The problem at hand

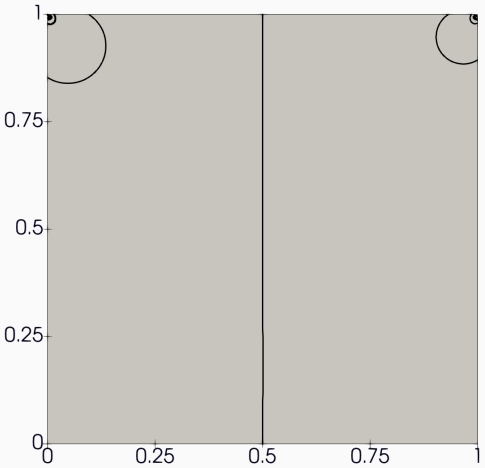
Studying the flow of a viscuous fluid in a 2D cavity



Expected resut



(a) Velocity streamlines



(b) Pressure isolines

Navier-Stokes equations

$$\begin{cases} \frac{\partial \mathbf{u}}{\partial t} - \operatorname{div}\left[\nu\left(\nabla \mathbf{u} + \nabla \mathbf{u}^T\right)\right] + (\mathbf{u}\cdot\nabla)\mathbf{u} + \nabla p = \mathbf{f}, & \mathbf{x} \in \Omega,\ t > 0, \\ \operatorname{div} \mathbf{u} = 0, & \mathbf{x} \in \Omega,\ t > 0, \end{cases}$$

Complete Incompressible Navier-Stokes System

Non-dimensionalisation

Units for the terms in the equation

$$[\nu] = \frac{L^2}{T}, \quad [u] = \frac{L}{T}, \quad [p] = \frac{L^2}{T^2} \quad \text{(kinematic pressure)}$$

Non-dimensionalisation

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Choose reference quantities

- characteristic length L
- characterisitc velocity U

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Choose reference quantities

- characteristic length L
- characterisitc velocity U

Defining dimensionless variables

$$x = Lx^*, \quad t = \frac{L}{U}t^*, \quad u = Uu^*, \quad p = U^2 p^*$$

Non-dimensionalisation

Each term is transformed

$$\frac{\partial \mathbf{u}}{\partial t} = \frac{U^2}{L}\frac{\partial \mathbf{u}^*}{\partial t}$$

$$-\nu\Delta\mathbf{u} = -\frac{\nu U}{L^2}\Delta^*\mathbf{u}^*$$

$$(\mathbf{u}\cdot\nabla)\mathbf{u} = \frac{U^2}{L}(\mathbf{u}^*\cdot\nabla^*)\mathbf{u}^*$$

$$\nabla p = \frac{U^2}{L}\nabla^* p^*$$

Non-dimensionalisation

Plugging in the transformed terms and dividing by $\frac{U^2}{L}$ gives

$$\frac{\partial \mathbf{u}^*}{\partial t^*} - \frac{\nu}{UL}\Delta^* \mathbf{u}^* + (\mathbf{u}^* \cdot \nabla^*)\mathbf{u}^* + \nabla^* p^* = 0$$

Define

$$\mathrm{Re} = \frac{UL}{\nu}$$

hence

$$\frac{\partial \mathbf{u}^*}{\partial t^*} - \frac{1}{\mathrm{Re}}\Delta^* \mathbf{u}^* + (\mathbf{u}^* \cdot \nabla^*)\mathbf{u}^* + \nabla^* p^* = 0$$

Stokes regime Re $\to$ 0

As the fluid gets more viscuous (Re $\to$ 0) turbulence disappears thus time derivative can be neglected.

$$\begin{cases} -\Delta \mathbf{u} + \nabla p = 0, & \mathbf{x} \in \Omega \\ \operatorname{div} \mathbf{u} = 0, & \mathbf{x} \in \Omega \end{cases}$$

$$\mathbf{u} = \mathbf{0}, \qquad \mathbf{x} \in \partial\Omega \setminus \Gamma,$$

$$\mathbf{u} = (1, 0)^T, \qquad \mathbf{x} \in \Gamma,$$

where $\Gamma = \{\mathbf{x} = (x_1, x_2)^T \in \partial\Omega : x_2 = 1\}$

Weak formulation of the Stokes problem

Find $(\mathbf{u}, p) \in V \times Q$ such that

$$\begin{cases} a(\mathbf{u}, \mathbf{v}) + b(\mathbf{v}, p) = 0, & \forall \mathbf{v} \in V \\ b(\mathbf{u}, q) = 0, & \forall q \in Q \end{cases}$$

where

$$a(\mathbf{u}, \mathbf{v}) = \int_\Omega \nabla \mathbf{u} \cdot \nabla \mathbf{v} \, d\Omega$$

$$b(\mathbf{u}, q) = -\int_\Omega q \, \mathrm{div}\, \mathbf{u} \, d\Omega$$

Appropriate spaces for V and Q

Let $\mathbf{v} \in V$ and $q \in Q$

$$V = \{\mathbf{v} \in \left[\mathrm{H}^1(\Omega)\right]^2 : \mathbf{v}|_{\Gamma_D} = 0\}$$

$$Q = \{q \in \mathrm{L}^2(\Omega) : \int_\Omega q \, d\Omega = 0\}$$

$$\int_\Omega \nabla \mathbf{u} \cdot \nabla \mathbf{v} \, d\Omega - \int_\Omega p \, \mathrm{div}\, \mathbf{v} \, d\Omega = 0$$

$$\int_\Omega (\mathrm{div}\, \mathbf{u}) \, q \, d\Omega = 0$$

Galerkin approximation

Let $V_h \subset V$, $Q_h \subset Q$ be finite dimensional subspaces. Find $u_h = \sum_{i=0}^{N_h} u_i \varphi_i$, $p_h = \sum_{k=0}^{M_h} p_k \phi_k$.

The Stokes problem becomes

$$\begin{cases} \mathrm{A}\mathbf{U} + \mathrm{B}^T\mathbf{P} = 0 \\ \mathrm{B}\mathbf{U} = 0 \end{cases}$$

$$\mathrm{A} = [a_{ij}] = [a(\varphi_j, \varphi_i)], \quad \mathrm{B} = [b_{km}] = [b(\varphi_m, \phi_k)]$$

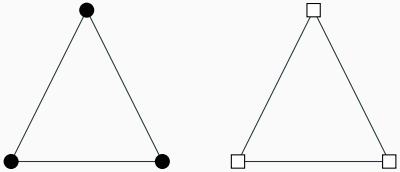
The inf-sup condition

There exists $\beta_h > 0$ such that

$$\forall q_h \in Q_h \; \exists v_h \in V_h, \; v_h \neq 0 \; : \; b(v_h, q_h) \geq \beta_h \|v_h\|_V \|q_h\|_Q$$

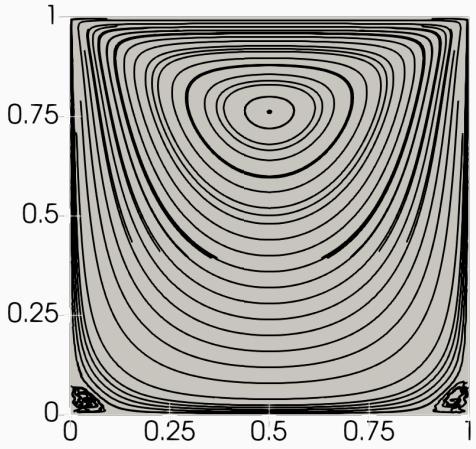
Unstable P1-P1 elements

$V = \mathbb{P}_1$, $Q = \mathbb{P}_1$ - continuous, piecewise polynomial functions

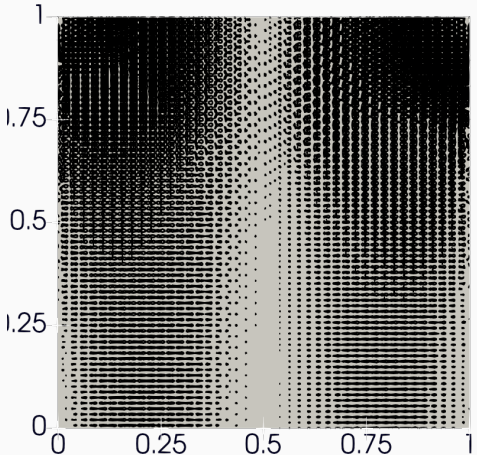


• represents DOFs for velocity and □ represents DOFs for pressure

Unstable P1-P1 elements



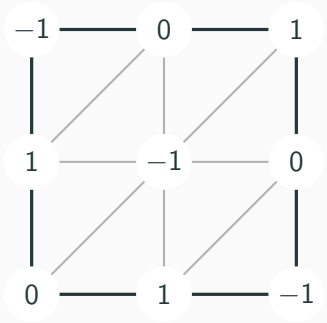
(a) Velocity streamlines



(b) Pressure isolines

Constructed pressure modes for P1-P1

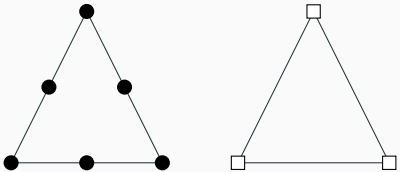
Assume that $(\mathbf{u_h}, p_h)$ is a solution of the Stokes problem, and $(\mathbf{u_h}, q_h^*)$ is as follows:



Then $(\mathbf{u_h}, p_h + q_h^*)$ is a solution with a spurious pressure mode.

Stable P2-P1 elements

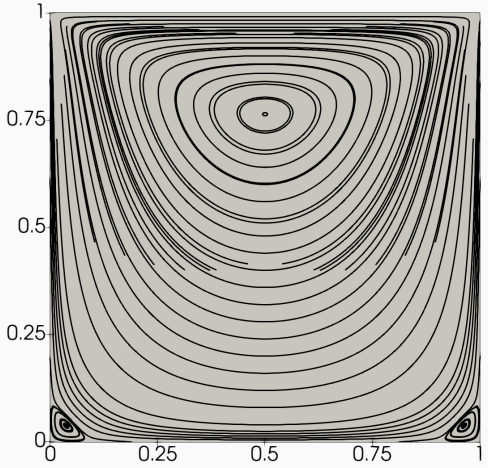
$V = \mathbb{P}_2$, $Q = \mathbb{P}_1$ - any $\mathbb{P}_k, \mathbb{P}_{k-1}$ for $k \geq 2$ is stable



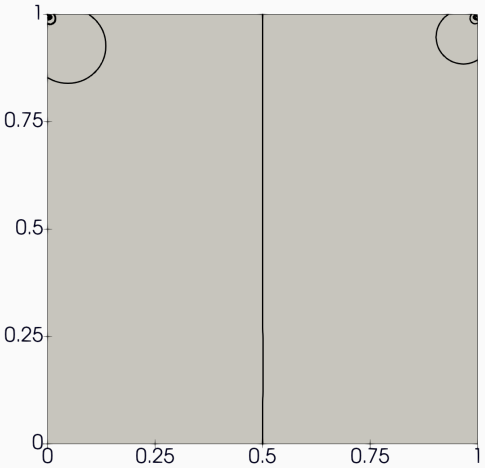
- represents DOFs for velocity and $\square$ represents DOFs for pressure

Proof of stability for $\mathbb{P}_k, \mathbb{P}_{k-1}$ is given in Brezzi & Fortin, Mixed and Hybrid Finite Element Methods, Springer, 1991

Stable P2-P1 elements



(a) Velocity streamlines

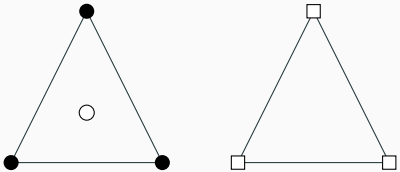


(b) Pressure isolines

P1-P1 stabilisation

Enrich $\mathbb{P}_1$ space with cubic bubble functions. $V = \mathbb{P}_1 \oplus B$ where

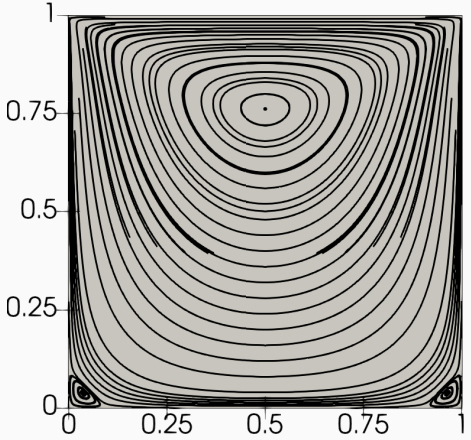
$$B = \{p \in \mathbb{P}_3 \,:\, p = 0 \quad \text{on the boundary}\}$$



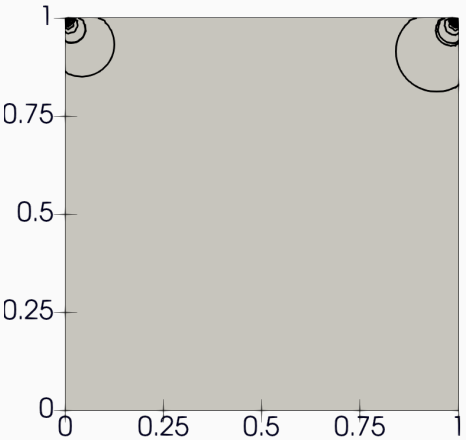
• velocity DOFs, ∘ bubble function DOFs, □ pressure DOFs

Stability of the bubble elements is shown in D. N. Arnold, F. Brezzi & M. Fortin (1984), "A stable finite element for the Stokes equations"

P1-P1 stabilisation



(a) Velocity streamlines



(b) Pressure isolines

Usefullness of AI

- research: valuable suggestions
- checking ones reasoning: valuable
- niche questions: unsatisfactory
- validation data: fabulations
- claude vs chatgpt vs u:ai

Thank you for your attention!